

MAT1002 Lecture 2 ,

Outline

- Infinite Sequences (10.1)
 - ↳ Bounded monotonic sequences
- Infinite Series (10.2)
 - ↳ Definitions
 - ↳ Properties of series
 - ↳ The n th-term test
 - ↳ Integral test (10.3)
 - ↳ Approximation

Bounded Monotonic Sequences

DEFINITIONS A sequence $\{a_n\}$ is **nondecreasing** if $a_n \leq a_{n+1}$ for all n . That is, $a_1 \leq a_2 \leq a_3 \leq \dots$. The sequence is **nonincreasing** if $a_n \geq a_{n+1}$ for all n . The sequence $\{a_n\}$ is **monotonic** if it is either nondecreasing or nonincreasing.

or monotone

- e.g. • Any constant sequence is both nondecreasing and nonincreasing.
• $\{(-1)^n\}$ is not monotonic.

10.1.6

Another name: Monotone Convergence Theorem

THEOREM 10.1.6 — The Monotonic Sequence Theorem If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

Proof: • Suppose $\{a_n\}$ is nondecreasing. (The other case: exercise.)

• Since $\{a_n\}$ is bounded, it is bounded above. Let M be the least upper bound for $\{a_n\}$. Will show that $a_n \rightarrow M$ as $n \rightarrow \infty$.

• Let $\varepsilon > 0$ be arbitrary. Then $M - \varepsilon$ is not an upper bound for $\{a_n\}$ (since M is the least), so $\exists N$ s.t. $a_N > M - \varepsilon$.

• Since $\{a_n\}$ is nondecreasing, $a_n \geq a_N > M - \varepsilon \quad \forall n > N$.

• Also, $a_n \leq M, \quad \forall n$.

• Therefore, $\forall n > N, \quad M - \varepsilon < a_n \leq M, \quad \text{so} \quad |a_n - M| < \varepsilon$.

• By def, $\lim_{n \rightarrow \infty} a_n = M$.



Infinite Series

Consider “adding” infinitely many numbers of the form $1/2^n$:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n} + \dots$$

One might argue that, intuitively, this “sum” is equal to 2.

On the other hand, “adding” infinitely many numbers does not always give a number: intuitively, the “sum” $1 + 1 + 1 + 1 + \dots$ tends to infinity.

Such “infinite sums” are formally understood as series.

How can we formalize the idea above? Consider

$$S_1 = 1, \quad S_2 = 1+1, \quad S_3 = 1+1+1, \quad \dots, \quad S_n = \underbrace{1+1+\dots+1}_{n \text{ times}} = n$$

$$\lim_{n \rightarrow \infty} S_n = \infty, \quad \text{so} \quad 1+1+1+1+\dots = \infty.$$

Definition

Given a sequence $\{a_n\}$ of real numbers ($n \geq 1$), define

$$s_n := \sum_{k=1}^n a_k \quad \text{and} \quad \sum_{n=1}^{\infty} a_n := \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k.$$

The symbol $\sum_{n=1}^{\infty} a_n$ is called an **infinite series**, or simply a **series**, of the sequence $\{a_n\}$. The number a_n is called the **n -th term** of the series, and the number s_n is called the **n -th partial sum**.

The series is said to be **convergent** if the limit exists, and is said to be **divergent** otherwise. (as a real number)

Remark

(or $\sum_n a_n$)

It is common to use $\sum_n a_n$ or $\sum a_n$ to denote a series for short.
Also, sometimes the initial index is an integer other than 1.

↳ e.g. $\sum_{n=89}^{\infty} a_n$

e.g. (a) $a_n \equiv a$ (constant sequence). Then

$$\sum_{n=1}^{\infty} a = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \underbrace{a + \dots + a}_{n \text{ times}} = \lim_{n \rightarrow \infty} na = \begin{cases} 0, & \text{if } a=0 \\ \infty, & \text{if } a>0 \\ -\infty, & \text{if } a<0 \end{cases}$$

Series is divergent

(b) Given any constant a and r (with $a \neq 0$), the series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots$$

is called the **geometric series**. If $r \neq 1$, then

$$S_n = a(1 + r + \dots + r^{n-1}) = a \frac{1-r^n}{1-r},$$

$$\text{So } \lim_{n \rightarrow \infty} S_n = \begin{cases} \frac{a}{1-r}, & \text{if } |r| < 1 \\ \text{D.N.E.}, & \text{if } |r| \geq 1 \end{cases}. \quad \text{If } r=1, \text{ then}$$

$$\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=1}^{\infty} a, \text{ diverges. Hence, the geometric series}$$

$$\sum_{n=1}^{\infty} ar^{n-1} \text{ converges if } |r| < 1, \text{ and diverges if } |r| \geq 1.$$

(c) For the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$, since $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$,

$$S_n = (1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots + (\frac{1}{n} - \frac{1}{n+1}) = 1 - \frac{1}{n+1},$$

$$\text{So } \sum_n \frac{1}{n(n+1)} = \lim_{n \rightarrow \infty} S_n = 1.$$

Remark

- A series can be written in many ways by index shifting:

$$a_1 + a_2 + \dots = \sum_{n=1}^{\infty} a_n = \sum_{n=0}^{\infty} a_{n+1} = \sum_{n=k}^{\infty} a_{n-k+1}$$

- Removing finitely many terms from a series does not affect its convergence.

Properties of Series

5.2.8

THEOREM 5.2.8 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. Sum Rule: $\sum (a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum (a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum k a_n = k \sum a_n = kA$ (any number k).

Idea:

1. Note that the partial sum of $\sum (a_n + b_n)$ is

$$a_1 + b_1 + a_2 + b_2 + \dots + a_n + b_n = \underbrace{(a_1 + \dots + a_n)}_{\rightarrow A \text{ as } n \rightarrow \infty} + \underbrace{(b_1 + \dots + b_n)}_{\rightarrow B \text{ as } n \rightarrow \infty}.$$

$\rightarrow A \text{ as } n \rightarrow \infty \quad \rightarrow B \text{ as } n \rightarrow \infty$

2 & 3 are similar.

Remark:

- If $k \neq 0$ is a constant, then

$$\sum a_n \text{ diverges} \Leftrightarrow \sum k a_n \text{ diverges.}$$

- If $\sum a_n$ converges but $\sum b_n$ diverges, then $\sum (a_n + b_n)$ diverges.

Convergence Tests

Unlike geometric series, it is generally hard to find the exact value of a series. We will discuss various tests for testing whether a series converges or not without computing its value.

The n^{th} Term Test

Suppose that a series $\sum a_n$ converges to S . Then

$$\lim_{n \rightarrow \infty} s_n = S = \lim_{n \rightarrow \infty} s_{n-1}.$$

Since $a_n = s_n - s_{n-1}$, we have

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = S - S = 0.$$

This yields the following n -th term test.

Theorem (The n -th Term Test)

If a series $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$; in other words, if $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$ diverges.

Example

- (a) The series $\sum_{n=1}^{\infty} \frac{e^n - 2}{4e^n + 5}$ diverges since $\lim_{n \rightarrow \infty} \frac{e^n - 2}{4e^n + 5} = \frac{1}{4} \neq 0$.
- (b) The series $\sum_{n=1}^{\infty} \sin \frac{n\pi}{2}$ diverges since $\lim_{n \rightarrow \infty} \sin \frac{n\pi}{2}$ does not exist.

Note that the converse of the n -th term test is false. That is, just having $a_n \rightarrow 0$ as $n \rightarrow \infty$ does not imply that $\sum a_n$ converges.

- (c) The series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

is called the **harmonic series**. Although $1/n \rightarrow 0$ as $n \rightarrow \infty$, this series is divergent.

Q: Why is the harmonic series divergent?

Series with Nonnegative Terms

Consider $\sum a_n$ with $a_n \geq 0 \forall n$. Then its partial sum sequence $\{S_n\}$ is nondecreasing, which converges if and only if it is bounded.

Theorem

Let $\{a_n\}$ be a sequence such that $a_n \geq 0$ for all n , and let $\{s_n\}$ be the corresponding sequence of partial sums. Then $\sum a_n$ converges if and only if $\{s_n\}$ is bounded.

Integral Test

Theorem (Integral Test)

($N \geq 1$)

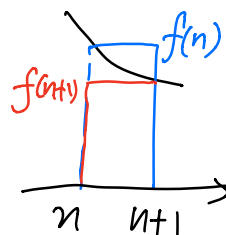
Suppose that $a_n = f(n) \geq 0$ for all n satisfying $n \geq N$ (with N being a fixed integer), where f is a nonincreasing continuous function on $[N, \infty)$. Then $\sum a_n$ converges if and only if the improper integral $\int_N^\infty f(x) dx$ converges.

Remark Note that the conditions above imply that $f(x) \geq 0$, $\forall x \in [N, \infty)$.

Proof: • Note that $\sum_{n=1}^\infty a_n$ converges if and only if $\sum_{n=N}^\infty a_n$ converges.

• For any $n \geq N$,

$$a_{n+1} = f(n+1) \leq \int_n^{n+1} f(x) dx \leq f(n) = a_n$$



• Hence

$$\sum_{n=n+1}^{n+k} a_n \leq \int_n^{n+k} f(x) dx \leq \sum_{n=N}^{n+k} a_n.$$

• Taking $k \rightarrow \infty$, we have

$$\sum_{n=n+1}^\infty a_n \leq \int_n^\infty f(x) dx \leq \sum_{n=N}^\infty a_n.$$

- If $\int_N^\infty f(x) dx$ converges, then the partial sums for $\sum_{n=N}^\infty a_n$ are bounded, so $\sum_{n=N}^\infty a_n$ (and hence $\sum_{n=1}^\infty a_n$) converges, by previous thm.
- If $\int_N^\infty f(x) dx$ diverges, then the partial sums for $\sum_{n=N}^\infty a_n$ are unbound, so $\sum_{n=N}^\infty a_n$ diverges, by previous theorem. $\square$

e.g. a) The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, where p is fixed, is called a **p-series**.

Case 1. If $p \leq 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} \neq 0$, so series diverges.

Case 2. If $p > 0$, then $f(x) = \frac{1}{x^p}$ is positive and decreasing on $(1, \infty)$.

Since $\int_1^{\infty} \frac{1}{x^p} dx$ converges for $p > 1$ and diverges for $0 < p \leq 1$ (done in MAT100).

By integral test on Case 2, together with Case 1, we have

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{converges, if } p > 1; \\ \text{diverges, if } p \leq 1. \end{cases}$$

In particular, the harmonic series ($p=1$) diverges.

b) Determine whether $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ is convergent.

Integral Test : Approximation

Suppose that we have a convergent series $\sum a_n$ satisfying the conditions in the integral test, and suppose that $\sum a_n = S$. Consider the error term $R_N := S - s_N$. Then

$$R_N = a_{N+1} + a_{N+2} + a_{N+3} + \dots = \sum_{n=N+1}^{\infty} a_n.$$

By the argument in the proof of the integral test, it follows that

$$\int_{N+1}^{\infty} f(x) dx \leq R_N \leq \int_N^{\infty} f(x) dx.$$

If we add s_N to the inequality above, we get

$$s_N + \int_{N+1}^{\infty} f(x) dx \leq S \leq s_N + \int_N^{\infty} f(x) dx. \quad (1)$$

This gives an interval I in which S lies, where $S = \sum a_n$. Hence, if we take the midpoint of I as an approximation of $\sum a_n$, the error is at most half the length of I , which is

$$\frac{1}{2} \left(\int_N^{\infty} f(x) dx - \int_{N+1}^{\infty} f(x) dx \right).$$

Note that this error is at most $a_N/2$. (Why?)

Example

Approximate the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using the method on the previous slide, with $N = 10$. Find an upper bound for the error.

Solution

- ▶ An easy computation shows that $\int_N^{\infty} (1/x^2) dx = 1/N$.
- ▶ By Inequality (1) on the previous slide, we have

$$s_{10} + \frac{1}{11} \leq S \leq s_{10} + \frac{1}{10}.$$

- ▶ $s_{10} = 1 + 1/4 + 1/9 + \dots + 1/100 \approx 1.54977$, so

$$1.64068 \leq S \leq 1.64977.$$

Then we take the midpoint of the interval, which is 1.645225. The error of this is at most $0.5 \times (1.64977 - 1.64068)$, which is 0.004545.

Fact: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} = 1.644934\dots$. Proof of this is not elementary.